Type MS · Kilobytes and kibibytes

Answer 1 of 3 · Question 2 part e · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part e in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

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Answer 1 of 3 · Question 2 part e · 2 marks

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Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

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Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

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Answer 1 of 3 · Question 2 part e · 2 marks

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Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

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Answer 1 of 3 · Question 2 part e · 2 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · Kilobytes and kibibytes

Answer 2 of 3 · Question 4 part a · 2 marks

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4(a)	Any **two** from: Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1) Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1) Kilobyte is equivalent to 10^3 bytes, whereas a kibibyte is equivalent to 2^{10} bytes (1) Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1)	Ignore units	2

Type MS · Kilobytes and kibibytes

Answer 2 of 3 · Question 4 part a · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **4 marks** for 19.17 or **One** mark for each of: • Calculates correct bits/bytes(1) • Adds metadata (1) • Calculates correct megabytes (1) • Completely correct response (1) Examples: $\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$ $\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$ ((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB ((2322 x 4128 x 2 bytes per colour) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB	• Arithmetic precedence rules must be clearly followed • Units are not required • Equivalent expressions are accepted • Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark	4

Type MS · Kilobytes and kibibytes

Answer 2 of 3 · Question 4 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Any **two** from: • To save disc/storage space (1) • Reduce transmission time (1) • Meet email attachment size restrictions (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Any **two** from: • Compressing / decompressing the file takes execution time (1) • Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1) • (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1) • Can affect the quality of the file (1) • File size can increase (1)		2

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Answer 2 of 3 · Question 4 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark
4(d)	Award 4 marks for 11 Award 3 marks for a value between 10 and 11 **One** mark for each: • 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1) • + 40 or + 80 (1) • ÷ 1024 (1) • Ceiling or RoundUp (1)	• Arithmetic precedence rules must be clearly followed • Units are not required • Alternative notations for ceiling are accepted • If 16 bits per Unicode character is not awarded, then allow follow through error • Allow follow through error, if interim calculations attempted • Equivalent expressions are accepted	4

Type MS · Kilobytes and kibibytes

Answer 2 of 3 · Question 4 part a · 2 marks

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Question Number	Answer	Mark

Type MS · Kilobytes and kibibytes

Answer 3 of 3 · Question 5 part b.iv · 1 mark

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5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

Type MS · Kilobytes and kibibytes

Answer 3 of 3 · Question 5 part b.iv · 1 mark

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Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

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Answer 3 of 3 · Question 5 part b.iv · 1 mark

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Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

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Answer 3 of 3 · Question 5 part b.iv · 1 mark

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Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 3 of 3 · Question 5 part b.iv · 1 mark

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Question	Answer	Additional Guidance	Mark